SHIVAM BHAGAT

Graduate Software Engineer · Python · TypeScript · PostgreSQL · CI/CD

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LinkedIn | GitHub | Portfolio | Right to work: UK Graduate visa (2 years)

PROFILE

MSc Advanced Computer Science graduate (University of Manchester, Dec 2026) seeking a graduate software engineer role. I build complete, working software rather than coursework demos: a web platform where authors publish and readers follow serialised novels, an Android reading app I use every day, and a developer tool that writes documentation for CI/CD pipelines, each with automated tests and CI/CD. Main stack: Python, TypeScript/React, PostgreSQL, GitHub Actions.

TECHNICAL SKILLS

Languages: Python (strongest), TypeScript, JavaScript, SQL, Kotlin (basic), C++ (academic)
Back end & data: REST APIs, PostgreSQL, Supabase, data modelling, row-level security, full-text search, YAML/JSON parsing
Front end & mobile: React, Vite, Tailwind CSS, Capacitor (Android), responsive UI
Testing & delivery: pytest, Vitest, regression testing, GitHub Actions CI/CD, Docker, Git, Linux, Vercel
Applied AI: LLM APIs (Gemini), retrieval-augmented generation, FAISS, LangGraph, Hugging Face
Practices: small reviewable pull requests, test-first delivery, documenting limitations, AI-assisted development (Claude Code)

PROJECTS

What Is My CI Pipeline Doing? – auto-documentation tool for CI/CD pipelines | GitHub — Feb – Sep 2026

MSc dissertation (solo) · Python · GitHub Actions · YAML · Mermaid · Gemini API · pytest

Problem: teams write docs for their build/deploy pipelines and the docs go out of date the moment the pipeline changes. This tool reads a repository's GitHub Actions workflows and generates the documentation itself: a Markdown fact sheet (triggers, jobs, dependencies, secrets, permissions) and a job-dependency diagram, with an optional AI-written overview on top.

- Designed the architecture so the AI can't make things up: a parser extracts the facts into a typed data model, 12 validation checks reject anything inconsistent, plain generators write the docs, and only then may Gemini rewrite the wording. Two commands: one workflow, or a whole repository with cross-workflow analysis.
- Engineered it like production software: **430+ automated tests**, regression tests on every CI run, a drift-check mode so CI fails when docs fall behind the pipeline, and a public LIMITATIONS file.
- Measured it on real open-source projects: against 201 pre-registered facts from 10 held-out workflows it captured **95.5% with zero false statements**; sending the raw YAML straight to the same LLM captured only 53.2%. No contradictions with the maintainers' own docs for scipy, tox and black.

StoryWanderer – web-novel reading & publishing platform | Live site — 2025 – present

solo full-stack project, deployed and in active development · React 18 · TypeScript · Vite · Tailwind CSS · shadcn/ui · Supabase (Auth, PostgreSQL) · Vercel

A website where authors publish serialised novels chapter by chapter and readers read and comment on them. Designed, built and deployed every layer solo.

- Front end: authentication, author dashboard, chapter editor with scheduled publishing, comments, custom long-form reader.
- Back end: PostgreSQL schema on Supabase with **row-level security** policies enforcing who can read and edit what, a privacy view for commenters, and full-text search with stored tsvector columns. Continuous deployment to Vercel.

Vach – offline PDF & EPUB reading app for Android | GitHub — Sep – Oct 2026

personal product, installed and used daily · React · TypeScript · Capacitor · Kotlin · pdf.js · Vitest · GitHub Actions

An Android reading app for people with a folder of ebooks on their phone. Pick the folder once and the app keeps access to it; every PDF and EPUB inside opens in one smooth continuous scroll with adjustable fonts, spacing and day/night themes. Works fully offline, with no ads, accounts or tracking.

- Hybrid architecture: React/TypeScript UI inside a Capacitor shell, plus a **Kotlin plugin I wrote** to use Android's Storage Access Framework, so folder permission persists across restarts and reboots instead of asking the user every time.
- Document pipeline: parses EPUB chapters (with HTML sanitisation for safety), extracts and reflows PDF text with pdf.js, falls back to page images for scanned PDFs, handles password-protected files, and remembers reading position per book.
- Delivery: GitHub Actions builds a debug APK on every push and a signed release on version tags; 10 Vitest suites cover parsing, sorting and position logic; verified on a physical device (Android back-button handling, error boundaries).

South Asian Culinary RAG Assistant – local AI recipe chatbot | GitHub — 2025 – 2026

MSc team project (4); owned retrieval · Python · LangGraph · FAISS · BGE-Large embeddings · Qwen2.5 · Hugging Face · Streamlit

- Built the search-and-routing core (FAISS vector search, BGE-Large embeddings, LangGraph intent router) of a chatbot that answers recipe questions from retrieved sources; **92.4% Recall@3** on a 100-query benchmark.

EDUCATION

MSc Advanced Computer Science | University of Manchester, UK — Sep 2025 – Sep 2026 · graduating Dec 2026

- Global Talent Scholarship recipient. Elected Academic Representative for the MSc cohort (student voice to faculty).
- Dissertation: built and evaluated a developer tool that auto-documents CI/CD pipelines (see Projects). Modules: Software Security, Security & Privacy in AI, NLP & Large Language Models, Topics in Machine Learning.

B.E. Computer Engineering, Honours in AI & Machine Learning | Savitribai Phule Pune University, India — 2021 – 2025

- First Class with Distinction: CGPA 8.35/10 (75.2%), equivalent to a UK First.

WORK EXPERIENCE (PART-TIME)

Tesco Extra, Altrincham | Online Order Picker (part-time) — Sep 2026 – present

- Pick and check online grocery orders accurately at pace; trace missing stock and escalate correctly; help customers on the shop floor.

Subway, Manchester | Team Member (part-time, alongside MSc) — May – Oct 2026

- Prepared made-to-order food and ran the till in a three-person team during rush periods; stock rotation and opening/closing to food-safety standards.

LEADERSHIP, VOLUNTEERING & CERTIFICATION

- **Academic Representative, University of Manchester (2025–26)** – elected voice of the MSc cohort to faculty; raised and followed up on course issues.
- **Event Organiser, IEEE Student Club, SPPU (2022–24)** – co-organised AI/ML workshops and hackathons for 200+ attendees.
- **Volunteer, Saday Sadev NGO and National Service Scheme (2022–25)** – outreach content, fundraising support, community and blood-donation drives.
- **ISC2 Certified in Cybersecurity (CC)** – 2024.